Design and Implementation of a Context-Aware Virtual Private Database using Role-Based Access Control and Auditing

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1. Narrative Description of the Project

Databases are widely used to store sensitive and valuable information such as personal details, financial records, medical data, and organizational documents. As modern applications increasingly rely on cloud-based and shared database environments, ensuring that this data is accessed securely has become a major concern. Unauthorized access, accidental data leakage, and insider misuse pose serious risks to data confidentiality and integrity. Many existing systems rely on basic access control mechanisms that are often too broad and do not provide sufficient protection for sensitive data.

This project will aim to design and implement a *Context-Aware Virtual Private Database (VPD)* that will provide fine-grained access control at the database level. Instead of granting users access to entire tables or datasets, the proposed system will restrict access to specific rows of data based on the user's role and current context. Contextual factors such as the user's role, time of access, and access environment will be considered while determining what data should be visible to a user. By combining Role-Based Access Control (RBAC) with context-aware security policies, the system will ensure that users will only be able to view or modify data that they are authorized to access under specific conditions.

The project will also include an auditing mechanism that will record all database access activities. Every attempt to read or modify data will be logged along with details such as the user identity, time of access, and type of operation performed. These audit logs will help in monitoring database usage, detecting suspicious behavior, and supporting post-incident analysis. By enforcing security policies directly at the database level and maintaining detailed audit records, the proposed solution will offer a more secure and reliable approach to protecting sensitive data in modern database systems.

Problem Statement. In many real-world database applications, access control is implemented mainly at the application layer rather than within the database itself. This approach leaves the database vulnerable to direct access, misconfigurations, and insider threats. Traditional access control models are often static and do not consider dynamic contextual factors such as when and under what conditions a user is accessing the data. As a result, users may gain access to more data than necessary, increasing the risk of data misuse or leakage. Furthermore, the lack of effective auditing mechanisms makes it difficult to track user actions, detect security violations, and investigate incidents. The problem addressed by this project is the absence of fine-grained, context-aware access control combined with reliable auditing in conventional database systems. The proposed system seeks to overcome these limitations by enforcing access control policies at the database level while maintaining comprehensive audit logs.

3. Identification of Information Needs

To successfully design and implement the proposed system, the following information is required:

- User identity and authentication credentials
- Role definitions and role-to-user mappings
- Contextual parameters such as time of access and access conditions
- Database records along with associated sensitivity levels
- Access control policies defining permitted and restricted operations
- Audit information including user actions, timestamps, accessed data, and access outcomes

This information enables precise policy enforcement, context evaluation, and effective auditing of database activities.

4. Distribution of Duties

The project responsibilities are divided among team members as follows:

- **Member 1 – Database Design and Security Implementation:** Will be responsible for database schema design, implementation of row-level security policies, role-based access control mechanisms, and configuration of auditing and logging features at the database level.
- **Member 2 – Application Development and System Integration:** Will be responsible for backend application development, user authentication and authorization, context evaluation logic, API design, and seamless integration of application components with the database security framework.

Both members will collaboratively contribute to system design, testing, documentation, and final project demonstration.